# Module 9.2 (Food Production)

**Food Security**

- when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food that meets their dietary needs and food preferences for an active and healthy life

**Factors Influencing Food Supply**

- poverty, war,
- bad weather
- climate change
- effects of industrialized food production

**Poverty** Is the Root Cause of Food Insecurity

**Malnutrition**

- Low income, less developed countries
- Diets center on high carbohydrate grains and very little protein
- No access to fresh food

**How food is produced**

- High Input Industrialized
- Low Input Traditional

**High Input Industrialized**

- Heavy equipment, fossil fuels, commercial fertilizer/pesticides,and money
- monoculture:growing one to two crops
- Food Supply vulnerable to disease

**Low Input Traditional**

- Solar energy and human labor to grow a crop that will feed a family with no surplus

**Organic Agriculture**

- crops grown without the use of synthetic pesticides and inorganic fertilizers, or genetic engineering

**Green Evolution**

- higher yields from existing cropland
    - Plant monocultures of selectively bred crops
    - Large amounts of water; synthetic fertilizers and pesticides
    - Multiple cropping

**Increasing Demand = Increase in Production**

**Gene Splicing**

- New and hardier crop varieties are being developed by second gene revolution methodology

*Meat production now uses feedlots as well as rangelands and pastures*

*Aquaculture produces more fish and is the world's fastest growing type of food production*

Industrialized Food Production **Requires Huge Inputs of Energy**

**Environmental Problems Arise from Industrialized Food Production**

- Soil Erosion
- Desertification
- Irrigation
- Water Shortages
- Air and Water Pollution
- Climate Change
- Loss of Biodiversity

*Industrialized agriculture allows farmers to use less land to produce more food but it is unsustainable*

**Topsoil Erosion is a serious problem**

- A significant natural capital component because it stores water and nutrients needed by plants
- Top soil recycle endlessly as long as they are not removed faster than natural processes replace them

**Soil Erosion**

- The movement of soil from one place to another by nature and by human activity

**Causes of Soil Erosion**

- Flowing water
- Wind loosens
- Farming
- Deforestation
- Grazing

**Harmful Effects of Soil Erosion**

- Loss of soil fertility (through depletion of plant nutrients)
- Kills fish and clog reservoirs and lakes
- Releases soil's carbon content

**Agricultural Activities Have Serious Environmental Consequences**

- Irrigation boosts/lowers farm productivity
- Irrigation water has a variety of salts, which leads to soil salinization
- Livestock production generates 18% of all greenhouse gas
- Fertilizer use releases nitrous oxide, increasing atmospheric temperatures

**Soil Salinization**

- Increased salt content in the soil

**Biodiversity Loss during food production**

- Cleaning and burning forests leads to loss of natural biodiversity
- Deforestation

**Limits to the expansion of Green Revolutions**

- Genetically Modified Food or GMO
- Genetically Engineered crop yields
- Population Growth and Water Availability\
- Deforestation speeding up climate change

**Harmful Environmental Effects of Industrialized Meat Production/Aquaculture**

- Cheap Meat produced by industrialized agriculture (causes health problems)
- Fishmeal and fish oil and food sources for farmed fish (depletes wild fish population)

## How Can We Protect Crops from Pests More Sustainably?
- By using a mix of cultivation techniques
- Using biological pest controls
- Using selective chemical pesticides

*Natural enemies control the populations of most pest species*

## Advantages of Synthetic Pesticide Use
- Human lives have been saved from insect transmitted disease (malaria)
- Food supplies are increased
- Crop yields and farming profits increased
- Newer pesticides are safer and more effective

## Disadvantages of Synthetic Pesticide Use
- Pesticides resistance in pest organisms
- Diminishes effectiveness - costing farmers more
- Kills off the pest's natural enemies
- Inefficient application

## Protective Laws and Treaties
## Agencies that regulate pesticides
- Environmental Protection Agency (EPA),
- U.S. Department of Agriculture (USDA)
- Food and Drug Administration (FDA) under the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA)

## Alternatives to Synthetic Pesticide Use
- Crop rotation/adjusting planting time
- Polyculture provides homes for pests' enemies
- Genetic resistance
- Biological Control (Use natural enemies)
    - Natural pheromones (insect perfume)
- Alteration of insect hormones
- Reducing synthetic herbicide to control weeds
- Integrated Pest Management (IPM) - coordinated combination of cultivation and chemical tools

## How to Improve Food Security
- Reduce Poverty and Malnutrition
- Producing food more sustainably
- Relying on locally sourced food

## The Government's Role in Improving Food Production and Security
- Controlling food prices vs. food subsides
- Implementing health measures
- Aid local, sustainable, organic food production and distribution
- Educate farmers
- Encourage Community Supported Agriculture (CSA) programs and vertical farming

## How Can We Produce Food More Sustainably?
- Efficient resource use
- Reducing the harmful effects of Industrialized food production
- Eliminating government subsidies

## Reducing Soil Erosion and Salinization, and Increasing Soil Fertility
- Soil Conservation
- Alley Cropping
- Use of Organic Fertilizer
- Reducing irrigation dependence

## Sustainable Food Production Solutions
- Open-ocean/polyaquaculture
- Re-circulating aquaculture systems
- Eat more chicken and eat less grain-efficient species (beef, pork and lamb)
- Eat more locally sourced, organic food, and have two meatless meals per week
- Switch to organic farming, perennial polyculture, renewable energy usage, and subsidies for sustainable food production

## Vertical Farming
- Vertical farming has a beneficial impact on the environment – especially in formerly industrialized areas
- can supply fresh food to urban food deserts at minimal cost – allowing people to have a healthier diet Vertical Farming and the Three Big Ideas

# MODULE 5.2: WATER RESOURCES AND WATER POLLUTION

**Freshwater**

- not evenly distributed over the earth's surface
- Globally, we have plenty but pollute and overuse it faster than natural processes can replenish it

**Freshwater Resources**

- Groundwater infiltrates downward through spaces in soil and rocks.

**Zone of saturation**

- underground areas of soil/rock where freshwater fills spaces between particles.

**Water Table**

- top of the groundwater zone; fluctuates up and down depending on weather, and removal/replenishment rates.

**Aquifer**

- underground body of rock that absorbs and holds flowing water.

**Surface Water**

- freshwater from rain and melted snow stored at the surface.
- 34% of the world's reliable surface runoff is used.

**Indirect and virtual water**

- used to produce food and other products this is a large part of our water footprint

**Freshwater Shortages**

- Water scarcity is caused by dry climate, drought, overuse/inefficient use, and using water faster than it can be replenished

**Freshwater Scarcity Stress**

- a calculation that compares freshwater availability with the amount used by humans.

**How Can We Increase Freshwater Supplies?**

- Groundwater for food production and use by cities is being pumped from aquifers faster than it can be replenished by nature
- Dam and reservoir systems and water transfer projects expand water distribution, they also disrupt ecosystems and displace people. Freshwater supplies can be augmented by the desalination of ocean water (expensive).

**Overpumping**

- limits food production, raises food prices, and widens the gap between the rich and the poor
- Can cause land subsidence and sinkhole development where land above aquifers collapses-making recharge impossible.

**Advantages of Large Dams and Water Transfer Projects**

- Dams/reservoir systems capture and store surface runoff from a river's watershed.
- Water is released as needed to control upstream flooding, generate electricity (hydropower), supply fresh/irrigation water, and provide recreational opportunities.
- Water transfer projects use dams, pumps, and aqueducts to transfer water from water-rich to water-poor regions.

**Disadvantages of Large Dams and Water Transfer Projects**

- Dam/reservoirs displace millions of people, flood productive lands, impair the ecosystem services of rivers, and have a useful life expectancy of only 50 years.
- Water transfer projects reduce a river's flow and flushing action (leading to pollution), and threaten fisheries and artificially cheapen costs leading to inefficient and wasteful water use

**How Can We Use Fresh Water More Sustainably?**

- By reducing wastage, raising prices, slowing population growth, and protecting ecosystems that store water naturally, we can use available freshwater more sustainably.

**Reducing Freshwater Losses and Improving Efficient Usage**

- More than half the world's freshwater supply is lost annually due to evaporation, and inefficient use (irrigation) because the government subsidies and underpricing and lack of subsidies for efficient water use. Switching to modern irrigation methods (drip, central pivot, etc.) will help reduce irrigation water usage by 10% – enough to supply everyone through 2025.

**Ways to Further Reduce Irrigation Water Losses**

**Solutions**

**Reducing Irrigation Water Losses**

- Avoid growing thirsty crops in dry areas
- Import water-intensive crops and meat
- Encourage organic farming and polyculture to retain soil moisture
- Monitor soil moisture to add water only when necessary
- Expand use of drip irrigation and other efficient methods
- Irrigate at night to reduce evaporation
- Line canals that bring water to irrigation ditches
- Irrigate with treated wastewater

- Flushing away industrial/household waste with freshwater causes pollution and is unsustainable.
- Gray water and industrial wastewater from sewage treatment plants can be used to clean equipment, flush away waste, water lawns, and irrigate non-food crops.

**Ways to Reduce Freshwater Losses in Industries**

**Solutions**

**Reducing Water Losses**

- Redesign manufacturing processes to use less water
- Recycle water in industry
- Fix water leaks
- Landscape yards with plants that require little water
- Use drip irrigation on gardens and lawns
- Use water-saving showerheads, faucets, appliances, and toilets (or waterless composting toilets)
- Collect and reuse gray water in and around houses, apartments, and office buildings
- Raise water prices and use meters, especially in dry urban areas

**Reduce your use and Waste of Freshwater**

## What Can You Do?

**Water Use and Waste**

- Use water-saving toilets, showerheads, and faucets
- Take short showers instead of baths
- Turn off sink faucets while brushing teeth, shaving, or washing
- Wash only full loads of clothes or use the lowest possible water-level setting for smaller loads
- Repair water leaks
- Wash your car from a bucket of soapy water, use gray water, and use the hose for rinsing only
- If you use a commercial car wash, try to find one that recycles its water
- Replace your lawn with native plants that need little if any watering
- Water lawns and gardens only in the early morning or evening and use gray water
- Use drip irrigation and mulch for gardens and flowerbeds

**How Can We Deal with Water Pollution?**

- Humans can use natural methods to treat sewage, cut resource use and waste, reduce poverty, and slow population growth to reduce water pollution – but the best way to reduce water pollution is to prevent it.

**Water pollution**

- is water quality changes that harm living organisms or make water unfit for drinking/ irrigation/recreation.

**Point sources**

- specific identifiable locations.

**Non-point sources**

- diffuse areas, difficult to identify/control, expensive to manage.

**Type of Water Pollution**

**Flowing streams and rivers**

- use a combination of dilution and bacterial biodegradation to naturally eliminate waste.
- Bacterial decomposition depletes dissolved oxygen eliminating populations of high oxygen using organisms until the stream is cleansed.

**Lakes and Reservoirs Pollution**

- have lower flow rates and are less effective at self-cleansing, as they typically have stratified layers.
- Nutrient overload produces algal/bacteria overgrowth, which depletes dissolved oxygen and kills off fish and marine organisms in bottom waters (dead zones).

**Groundwater Pollution**

- cannot cleanse itself of degradable wastes as quickly as flowing surface water.
- has lower concentrations of dissolved oxygen/smaller populations of decomposing bacteria.

**Ocean Pollution**

- Many humans treat the ocean as a dumping site. May be safer to dump wastes and degradable pollutants into the deep ocean, where it can be diluted/dispersed/degraded.

**Contaminants in the ocean are:**

- Viruses in raw sewage and from sewage treatment plants
- Toxic chemicals, garbage, sewage, and waste oil from cruise ships
- Nitrates/phosphates and sewage from agricultural waste
- Crude and refined petroleum
- Urban and industrial runoff

**Reducing Water Pollution**

- Reduce soil erosion by keeping cropland covered with vegetation, and by using conservation tillage.
- Use slow-release fertilizer – and no fertilizer on steeply sloped land.
- Plant vegetative buffer zones between cultivated fields and nearby surface waters.
- Encourage organic farming.

**Sewage Treatment Plants**

- Waterborne wastes in urban areas from homes, businesses, and storm runoff flow through pipes to sewage treatment plants.

**Primary sewage treatment**

- a physical/mechanical process

**Secondary sewage treatment**

- a biological process

**Sustainable Ways To Reduce and Prevent Water Pollution**

- Find substitutes for toxic pollutants
- Remove hazardous waste before it reaches sewage treatment facilities
- Use natural sewage treatment methods
- Reduce non-point runoff
- Slow population growth/reduce poverty
- Eliminate air pollution
- Encourage recycling/reuse of resources

**Net Energy Yield**

- is the amount of energy obtained from a resource minus the amount of energy needed to produce it
- Net energy yield values vary greatly depending on the source of energy - Energy input: energy needed to produce energy

**Net energy yield = total energy produced – energy required to produce it**

  - If a net energy yield is zero or a negative number. The resource cannot compete in the marketplace.

**Using Fossil Fuels**

- Humans use fossil fuels because they are available and have high energy content.
- renewable fuels degrade the environment, causes air and water pollution, and releases greenhouse gases into the atmosphere
- Fossil fuels produce most of our energy, but they are nonrenewable energy – violating the principle of sustainability
- Crude oil/petroleum: formed by pressure applied to decayed organic remains.
- Natural gas and crude oil are often found together in deposits.

**Extracting and Refining of Oil**

- Finding/extracting oil: 3-D seismic maps and computers to find deposits, drill to check deposits, then drill production wells.
- Crude oil must be refined to be usable reduces net energy yield.

**Natural gas**

- mixture of gases provides 28% of energy
- With a medium net energy yield used in cooking, heating and industrial purposes; cleaner than oil and coal.

**Liquefied petroleum gas (propane and butane)**
- tapped from deposits and stored in pressurized tanks.

**Coal**
- solid fossil fuel formed from decaying organic matter exposed to heat and pressure over millions of years
- **Coal is the dirtiest fossil fuel, polluting air and water**
- Coal is cheap due to low market pricing.

**Coal burning**
- power/industrial plants largest emitters of greenhouse gases
- this produces toxic coal ash.

**Peat**
- is a soil material made of moist, partially decomposed organic matter, similar to coal
- It is not classified as a coal, although it is used as a fuel.

**Nuclear Power**
- Nuclear power has little environmental impact and a very low accident risk, but usage is limited due to its low net energy yield.
- Nuclear fission chemical reactions provide the heat inside a reactor – process is complex and costly.

**Fuel**
- Is ***uranium*** ore contained in fuel rods and water as a coolant circulates through the reactor – reactor is surrounded by a steel containment shell.

**Nuclear fusion**
- when two lighter atoms are forced together at high temperatures to form a heavier atom, energy is released
- using controlled nuclear fusion will create almost limitless supplies of energy.
- may also be used for destroying toxic wastes and desalinization.

**Energy efficiency**
- measure of work from each unit of energy, meaning that we need more work for less energy.

**Improving Energy Efficiency**
- use a combined heat and power system to recycle steam as heat
- Make electric car motors more efficient
- Recycle materials, especially steel and other metals
- Improve designs of data centers
- Convert electrical grids into smart grids
- Connect solar and wind power to grids

**Energy Efficiency and Green Construction**

Use principles of sustainability

**Green architecture**
- solar heating, efficient windows, appliances and lighting

**Green roofs**
- soil and vegetation roofs that help insulate a building

**Superinsulation**
- air tight structures are heated/cooled mainly with sunlight, appliances and body heat.

**Renewable Energy**
- can meet our energy needs while reducing the effects on the environment less pollution, greenhouse gas emissions, and biodiversity loss.
- Heat buildings and water with solar energy

**Passive** (building absorbs heat directly)

**Active solar heating** (energy stored in rooftop solar collectors)

**Cool buildings**
- Plant trees for shade
- Use light colored roofs to reflect heat and geothermal heat pumps to pump cool air from underground

**Solar Power**
- Solar cells convert sunlight to electrical energy (no pollutants/greenhouse gases)
- May provide electricity to isolated areas of less-developed countries.
- Low to medium net energy yield, but efficiency technology is improving

**Hydropower**
- use of (kinetic energy) falling or flowing water to generate electricity

**Microhydropower generators**
- portable floating turbines that can use a stream or river for power without altering the environment.

**Wind Power**
- has the potential to produce 40x the current global use of electricity
- Onshore wind farms
- Offshore wind farms
- Even with full-cost pricing, windpower is the least costly way to produce electricity.

**Drawbacks**
- Regions with the highest sustainable winds are often sparsely populated/located far from cities so electrical grids would have to be replaced and expanded using smart grids

**Liquid Biofuels**
- Biomass can be burned as a solid fuel
- ***Ethanol*** (from plants and plant wastes) and ***Biodiesel*** (from vegetable oil) have advantages over gasoline

**Biofuel crops**
- grow anywhere and reduce dependence on imported oil

**Geothermal Energy**
- Heat stored in soil, underground rocks and fluids in the earth's mantle can be used to heat/cool buildings and produce electricity and captured by:
- *Geothermal heat pump systems*
- *Hydrothermal reservoirs of geothermal energy*
- *Hot, dry rocks deep underground*

**Transition To a More Sustainable Energy Future**
- Dramatically improve energy efficiency
- Use a mix of renewable energy resources
- Adjust market prices to include environmental and health costs